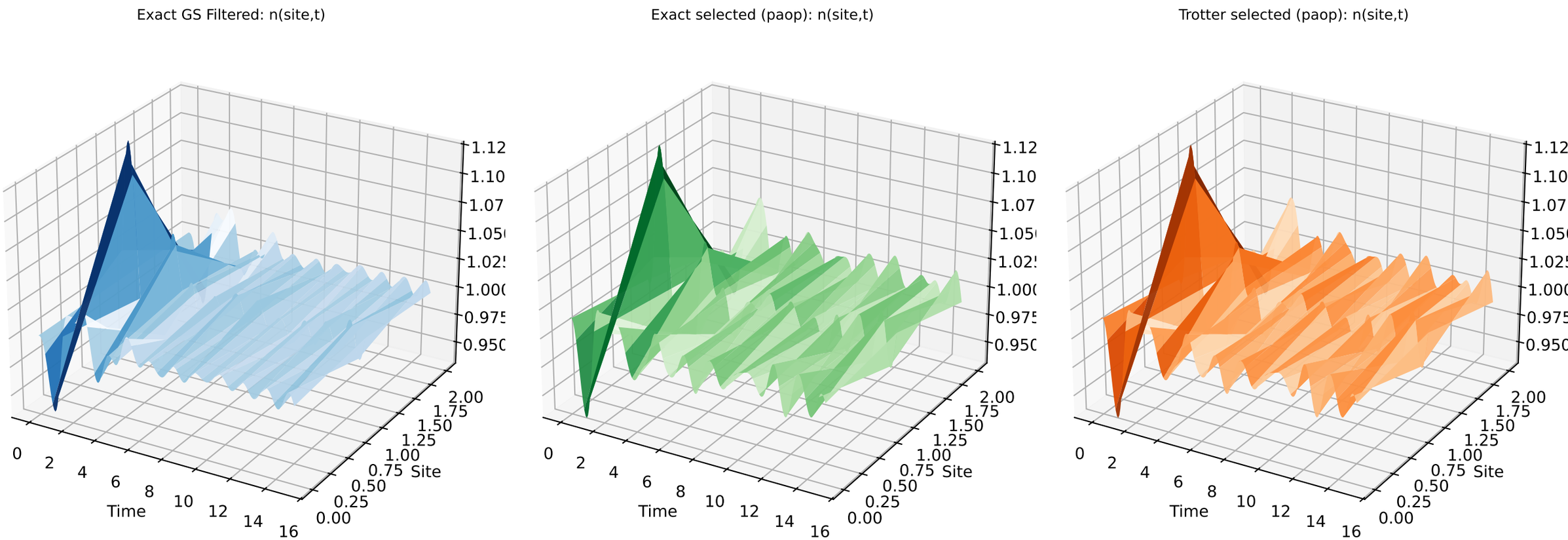
```
Parameter Manifest — Hubbard-Holstein L=3

Model:
  - model_family: Hubbard-Holstein
  - problem: hh

Ansatz:
  - ansatz_type: hh_hva_ptw
  - vqe_method: SPSA
  - vqe_method_internal: hardcoded_hh_hva_physical_termwise_statevector
  - initial_state_source: adapt_json
  - branch_order: exact_gs_filtered, exact_paop, trotter_paop, exact_hva, trotter_hva

Drive Enabled: True
  - spsa_a=0.2  spsa_c=0.1  spsa_A=10.0
  - spsa_alpha=0.602  spsa_gamma=0.101
  - spsa_eval_repeats=1  spsa_eval_agg=mean  spsa_avg_last=0

Branch Provenance:
  - paop_source: adapt_json
  - paop_pool: full_meta
  - paop_depth: None
  - hva_source: regular_vqe
  - hva_ansatz: hh_hva_ptw
  - hva_vqe_success: True

ADAPT Import:
  - source_json: L3_hh_smart_replay_bundle.json
  - metadata_match_passed: True
  - strict_match: True
  - pool: full_meta
  - ansatz_depth: None
  - adapt_energy: 0.2929667660817128
  - adapt_abs_delta_e: 0.04802606595379805
```

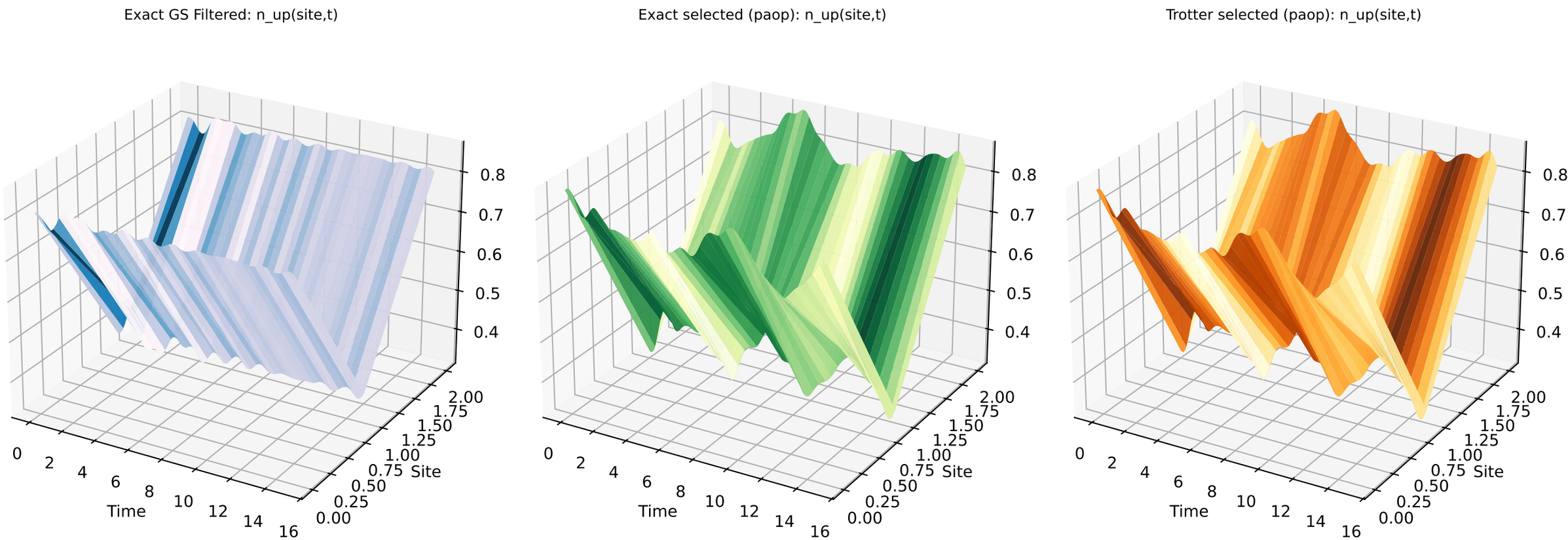
```
  - A=0.5  omega=2.0
  - tbar=3.0  phi=0.0  t0=0.0
  - pattern=staggered  time_sampling=midpoint
  - exact_steps_multiplier=2

Core Physical Parameters:
  - t=1.0
  - U=4.0
  - dv=0.0

Hubbard-Holstein Parameters:
  - omega0=1.0
  - g_ep=0.5
  - n_ph_max=1
  - boson_encoding=binary
  - nq_fermion=6
  - nq_phonon=3
  - nq_total=9

Trajectory Settings:
  - t_final=15.0
  - num_times=201
  - trotter_steps=192
  - suzuki_order=2
  - term_order=sorted
  - boundary=open
  - ordering=blocked
  - initial_state_source=adapt_json

Topline Results:
  - subspace_fidelity_at_t0: 0.9819573966530011
  - vqe_energy: 0.38444266279987005
  - exact_filtered_energy: 0.24494070012791275
  - qpe_energy_estimate: None
```

```
ADAPT Initial-State Provenance

import_json: L3_hh_smart_replay_bundle.json
import_source: hh_vqe_from_adapt_family.best_state
strict_match: True
metadata_match_passed: True

pool_type: full_meta
ansatz_depth: None
num_parameters: None
operator_count: None
adapt_energy: 0.2929667660817128
adapt_abs_delta_e: 0.04802606595379805

Dynamics used imported ADAPT state as t=0 initial condition.
```

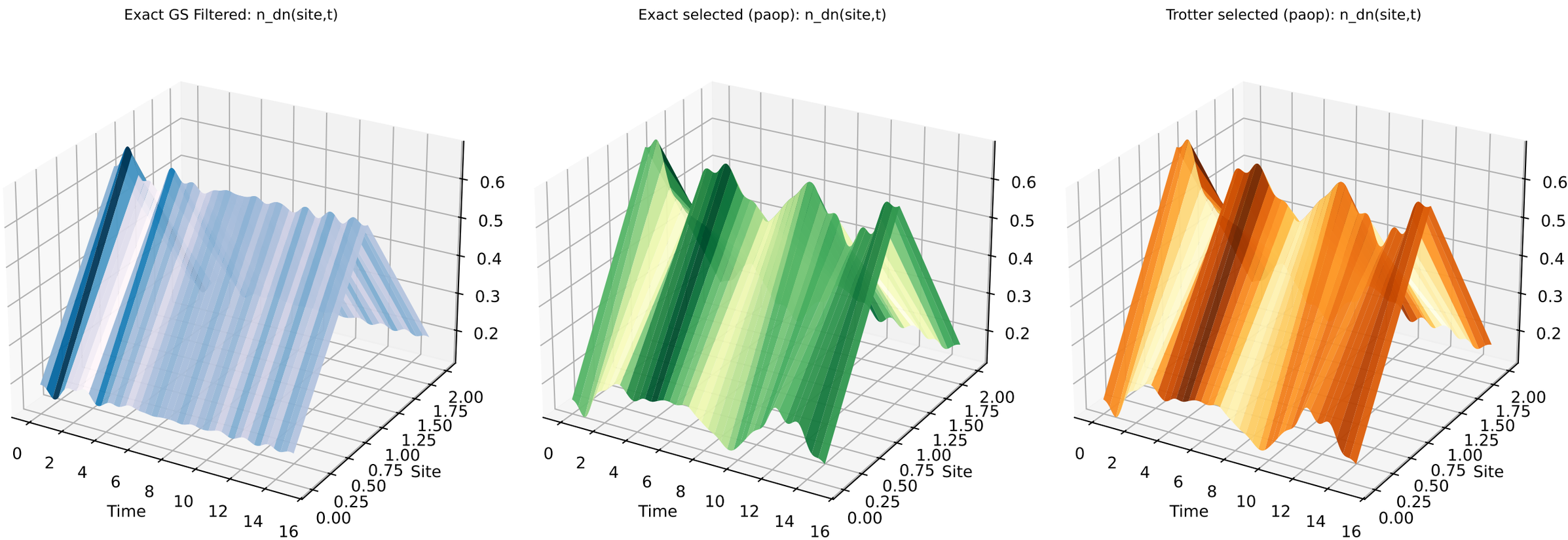
L=3 3D Densities (Total n)



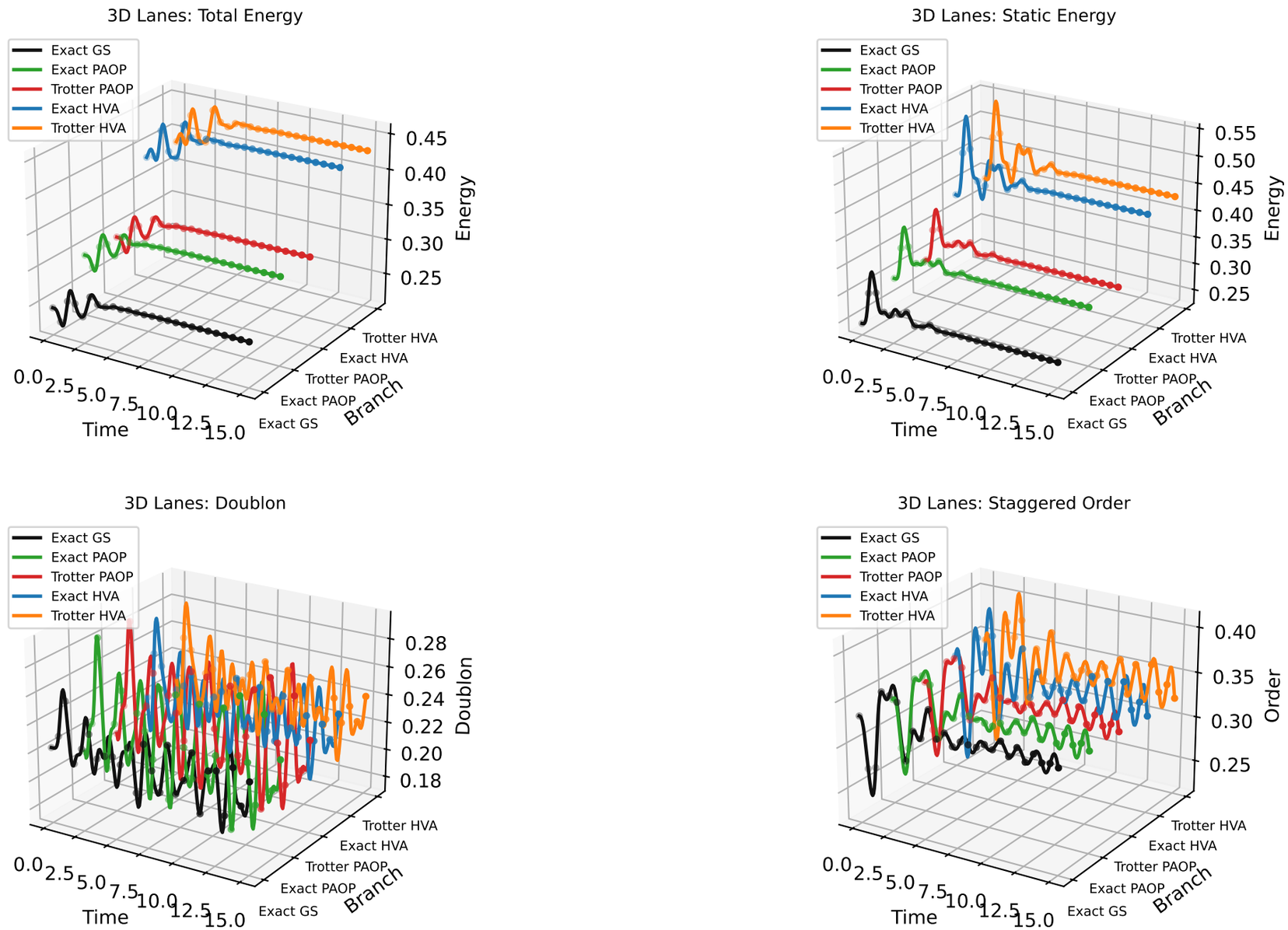
L=3 3D Densities (Spin-Up)

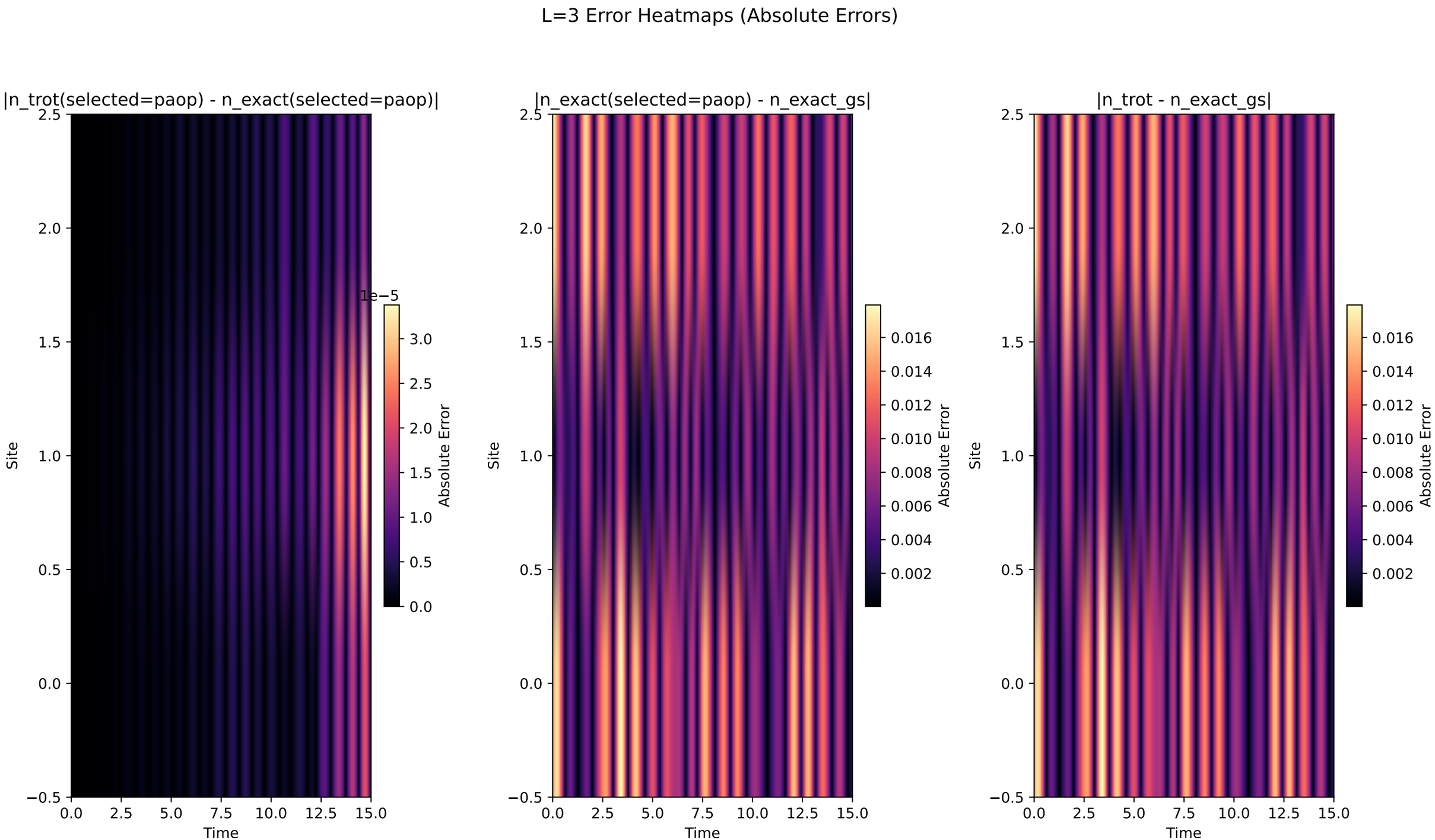


L=3 3D Densities (Spin-Down)

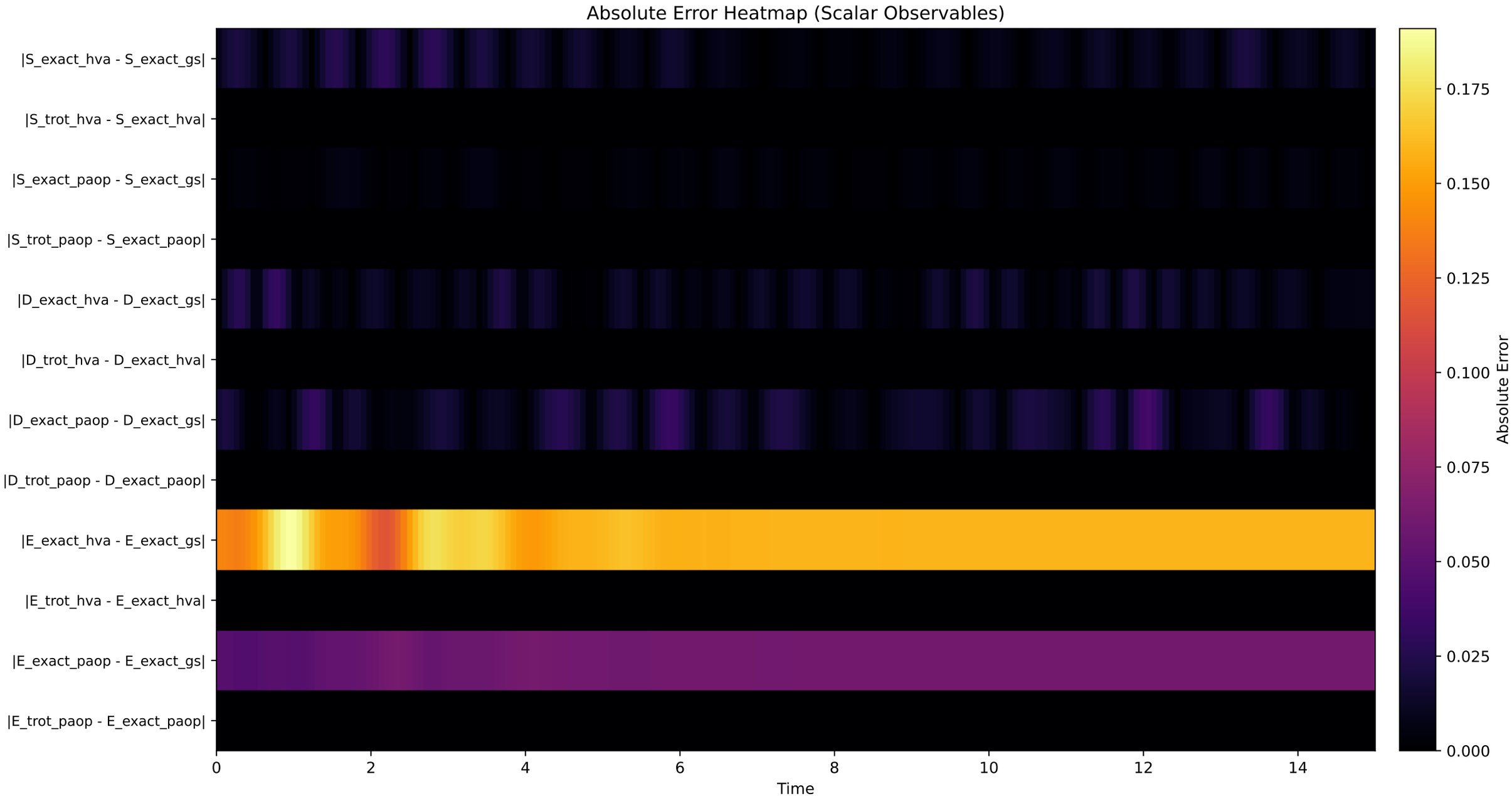


L=3 3D Scalar Observables (Five Branches)

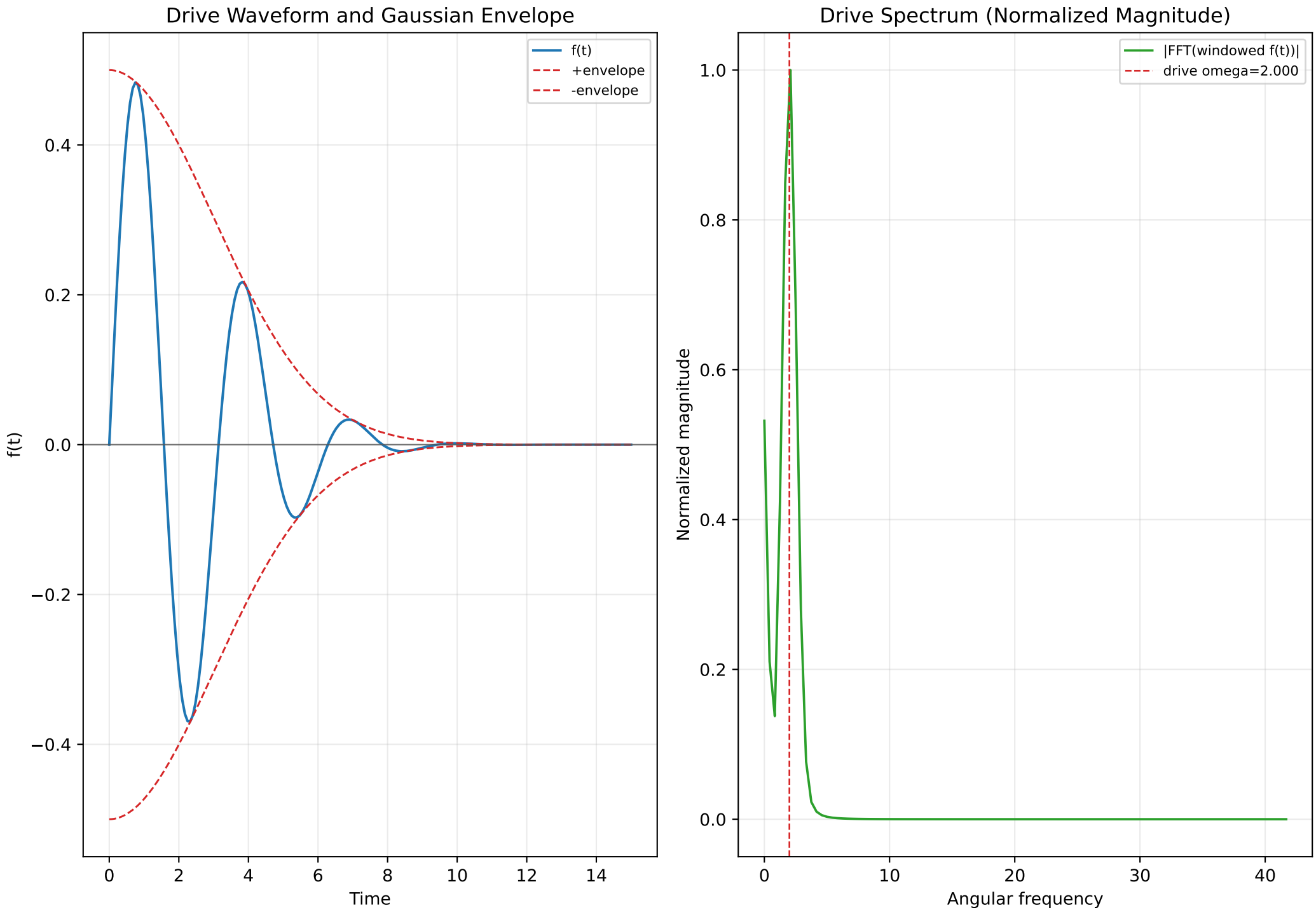



L=3 Error Heatmaps (Absolute Errors)
|n_trot(selected=paop) - n_exact(selected=paop)|
|n_exact(selected=paop) - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1e−5
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
3.0
0.016
0.014
0.012
0.010
0.008
0.006
0.004
0.002

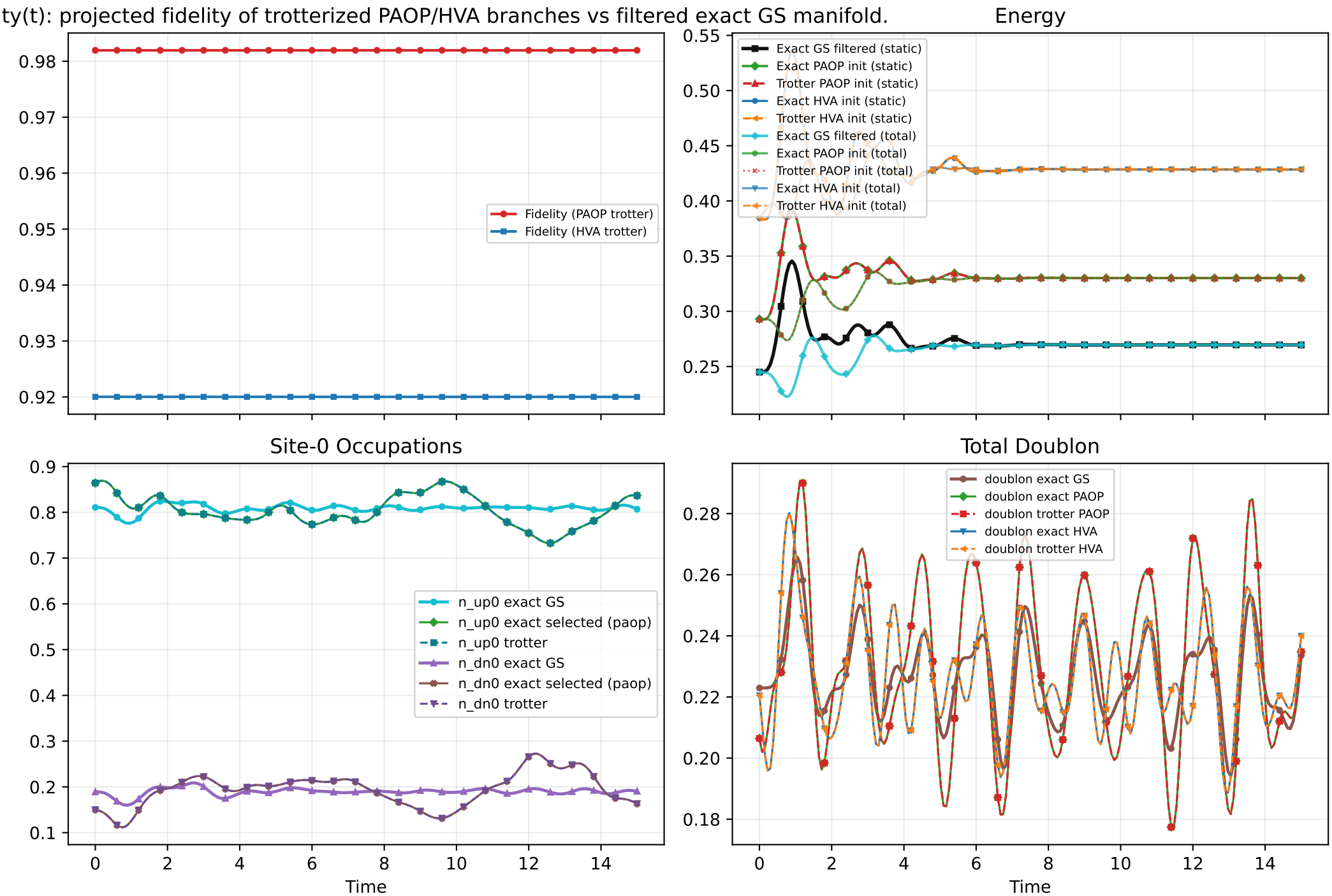
L=3 Scalar Error Heatmap

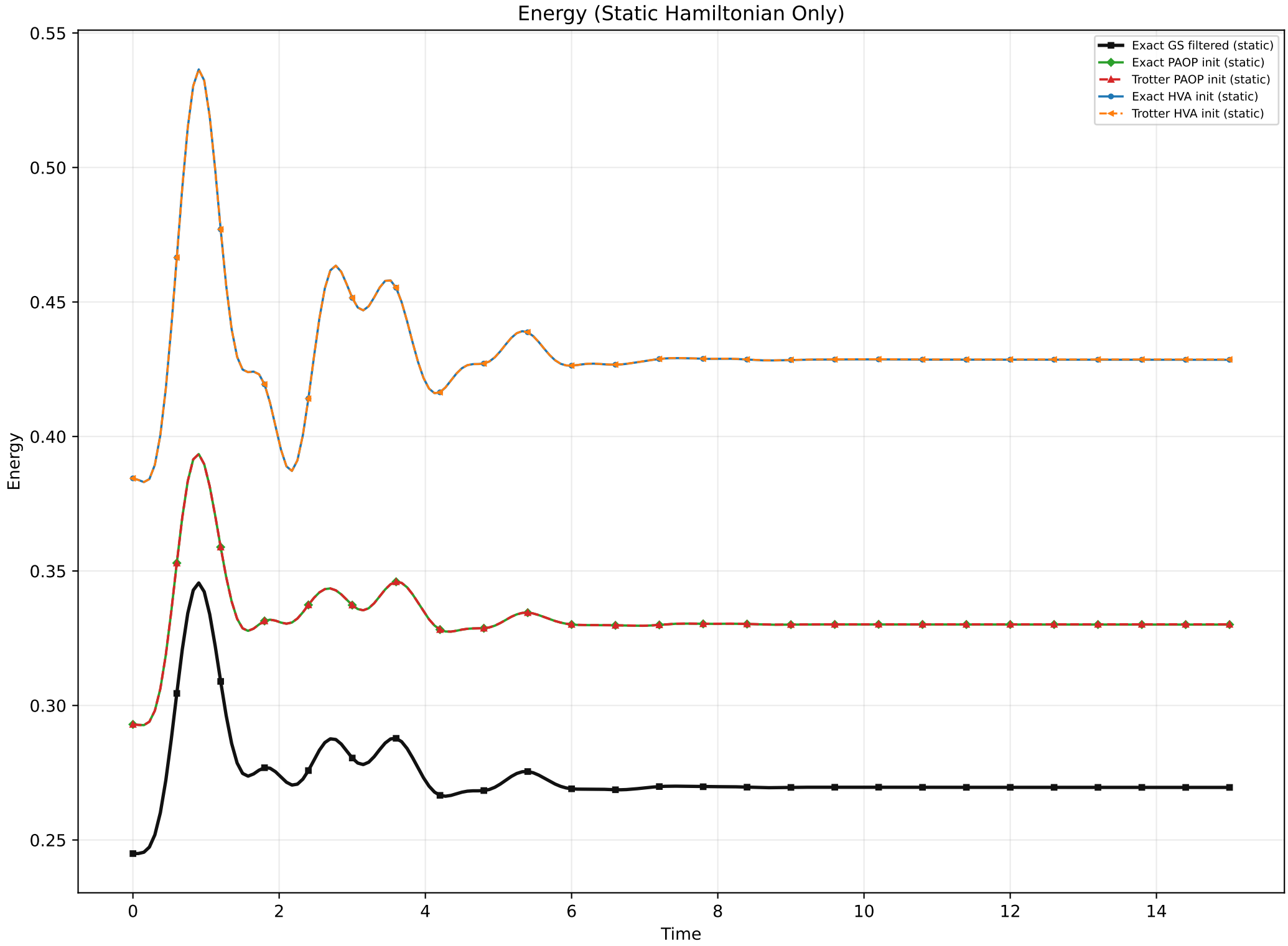


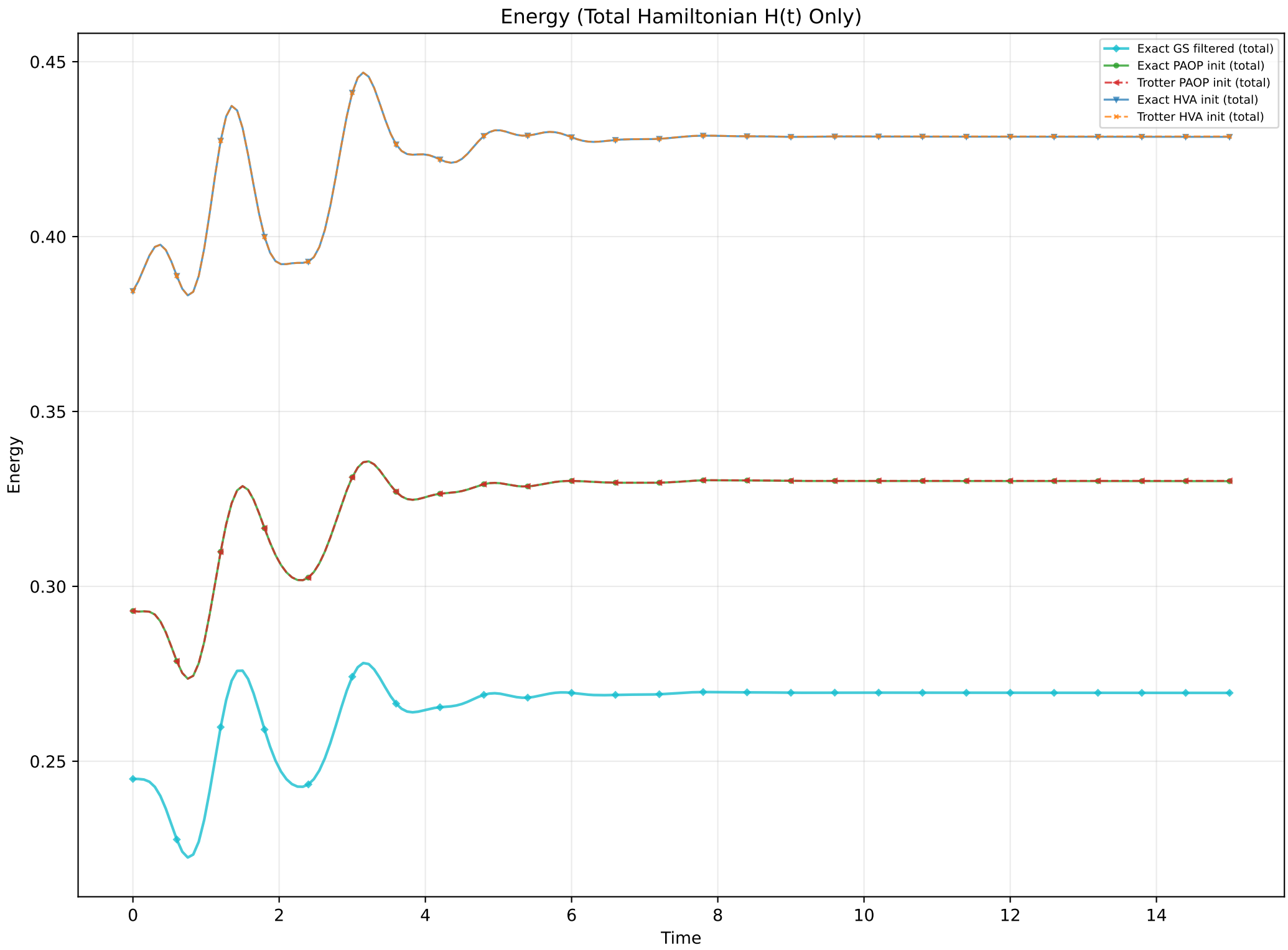
Drive Diagnostics: A=0.5, omega=2.0, tbar=3.0, phi=0.0, t0=0.0



Hardcoded Hubbard Pipeline: L=3



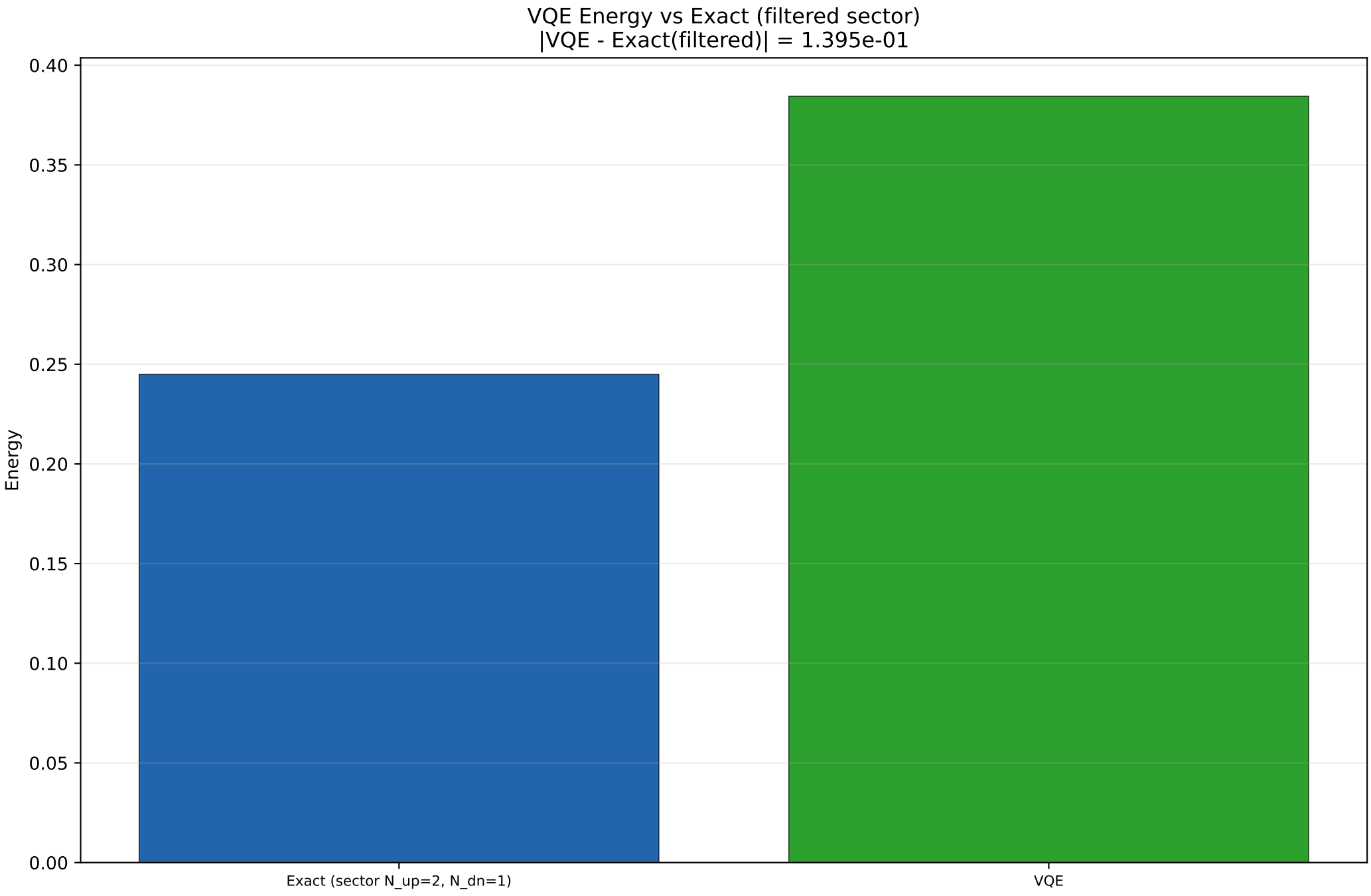

Energy (Static Hamiltonian Only)
Exact GS filtered (static)
Exact PAOP init (static)
Trotter PAOP init (static)
Exact HVA init (static)
Trotter HVA init (static)
Energy
Time
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0
2
4
6
8
10
12
14


Energy (Total Hamiltonian H(t) Only)
Exact GS filtered (total)
Exact PAOP init (total)
Trotter PAOP init (total)
Exact HVA init (total)
Trotter HVA init (total)
Energy
Time
0.45
0.40
0.35
0.30
0.25
0
2
4
6
8
10
12
14

Appendix — L=3

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

Ansatz:
  - hardcoded ansatz: hh_hva_ptw
  - vqe method: hardcoded_hh_hva_physical_termwise_statevector

Energy + Fidelity:
  - subspace_fidelity_at_t0: 0.9819573966530011
  - subspace_fidelity_paop_t0: 0.9819573966530011
  - subspace_fidelity_hva_t0: 0.9200154974625769
  - energy_t0_exact_gs: 0.24494070012791422
  - energy_t0_exact_paop: 0.29296676608171324
  - energy_t0_trotter_paop: 0.29296676608171324
  - energy_t0_exact_hva: 0.3844426627998696
  - energy_t0_trotter_hva: 0.3844426627998696
  - ground_state_exact_energy_full_hilbert: -0.638955801088
  - ground_state_exact_energy_filtered: 0.24494070012791275
  - filtered_sector: {'n_up': 2, 'n_dn': 1}
  - vqe_energy: 0.38444266279987005
  - qpe_energy_estimate: None

Config:
  - initial_state_source: adapt_json
  - exact_trajectory_label: Exact_Hardcode
  - exact_trajectory_method: python_matrix_eigendecomposition
  - fidelity_definition: fidelity_paop_trotter(t) = <psi_paop_trot(t)|P_exact_gs_subspace(t)|psi_paop_trot(t)> and
    fidelity_hva_trotter(t) = <psi_hva_trot(t)|P_exact_gs_subspace(t)|psi_hva_trot(t)>, where
    P_exact_gs_subspace(t) projects onto the time-evolved filtered-sector ground manifold selected by E <= E0 +
    tol. Legacy key fidelity(t) follows the selected initial-state branch.
  - hamiltonian_terms: 27
  - reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```